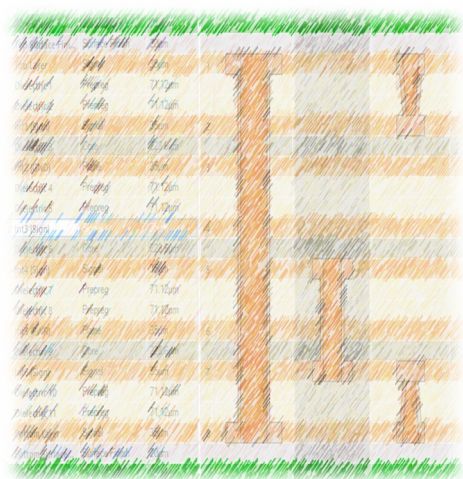


INTEGRIDAD DE SEÑAL

DISEÑO *STACK-UPS* - IMPEDANCIA DE LÍNEA

REGLAS DE DISEÑO *HIGH-SPEED* EN ALTIVM DESIGNER



José F. Toledo (jtoledo@eln.upv.es) y Germán Ramos Peinado (gramosp@eln.upv.es)

CONTENIDOS

- Impedancia de línea con *stack-up* editor
- Reglas de diseño básicas
- Reglas de diseño *High Speed*
- Reglas de diseño: directivas en esquemas

IMPEDANCIA DE LÍNEA CON STACK-UP EDITOR

1 Design Tools Route Reports Window Help

Update Schematics in Stackup_practice.PrjPcb
Import Changes From Stackup_practice.PrjPcb

Rules...
Rule Wizard...

Board Shape

Netlist

Signals

Layer Stack Manager...

Manage Layer Sets

Rooms

Classes...

Make PCB Library

Make Integrated Library

2 Tools Window Help

Layer Stack Visualizer

Material Library

Features

Presets

Measurement Units

Create Via Types From Used Vias

Stackup_practice1.PcbDoc [Stackup]

Delete

2 Layers (2 x Signal, 0 x Plane)
4 Layers (2 x Signal, 2 x Plane)
6 Layers (4 x Signal, 2 x Plane)
8 Layers (6 x Signal, 2 x Plane)
10 Layers (6 x Signal, 4 x Plane)
12 Layers (6 x Signal, 6 x Plane)
14 Layers (8 x Signal, 6 x Plane)
16 Layers (8 x Signal, 8 x Plane)

1 Top Layer
Dielectric 1 FR-4
2 Bottom Layer
Bottom Solder Solder Resist
Bottom Overlay

3 Tools Window Help

Layer Stack Visualizer

Material Library

Features

Presets

Measurement Units

Create Via Types From Used Vias

Surface Finish

Signal 1oz

Thickness

mil
in
um
mm

4

+ Add Remove Features

#	Name	Material	Type	Weight	Thickness	Dk	Top Ref	Bottom Ref	Width (W1)	Trace d
	Top Overlay		Overlay							
	Top Solder	SM-001	Solder Mask		25.4µm	4				
	Top Surface Fini...	PbSn	Surface Finish		20µm					
1	Top Layer	CF-004	Signal	1oz	35µm					
	Dielectric 1	PP-006	Prepreg		71.12µm	4.1				
	Dielectric 2	PP-006	Prepreg		71.12µm	4.1				
2	Int1 (Sign)	CF-004	Signal	1oz	35µm					
	Dielectric 3	Core-027	Core		228.6µm	4.5				
3	Int2 (GND)	CF-004	Plane	1oz	35µm					
	Dielectric 4	PP-006	Prepreg		71.12µm	4.1				

Stackup Impedance Via Types

Add Impedance Profile

IMPEDANCIA DE LÍNEA CON STACK-UP EDITOR

Stackup_practice1.PcbDoc * Stackup_practice1.PcbDoc [Stackup] *

+ Add Remove \$50 Features

#	Name	Material	Type	Weight	Thickness	Dk	Top Ref	Bottom Ref	Width (W1)	Impe...	Devia...	Delay...	
	Top Overlay		Overlay										
	Top Solder	SM-001	Solder Mask		25.4µm	4							
	Top Surface Fini...	PbSn	Surface Finish		20µm								
1	Top Layer	CF-004	Signal	1oz	35µm		✓	2 - Int1 (Sign)	224.37µm	49.99	0.02%	6.0745	
	Dielectric 1	PP-006	Prepreg		71.12µm	4.1							
	Dielectric 2	PP-006	Prepreg		71.12µm	4.1							
2	Int1 (Sign)	CF-004	Signal	1oz	35µm		✓	1 - Top Layer	3 - Int2 (GND)	124.69µm	49.99	0.01%	6.9403
	Dielectric 3	Core-027	Core		228.6µm	4.5							
3	Int2 (GND)	CF-004	Plane	1oz	35µm								
	Dielectric 4	PP-006	Prepreg		71.12µm	4.1							
	Dielectric 5	PP-006	Prepreg		71.12µm	4.1							
4	Int3 (Sign)	CF-004	Signal	1oz	35µm		✓	3 - Int2 (GND)	5 - Int4 (Sign)	124.69µm	49.99	0.01%	6.9403
	Dielectric 6	Core-027	Core		228.6µm	4.5							
5	Int4 (Sign)	CF-004	Signal	1oz	35µm		✓	4 - Int3 (Sign)	6 - Int5 (PWR)	124.69µm	49.99	0.01%	6.9403
	Dielectric 7	PP-006	Prepreg		71.12µm	4.1							
	Dielectric 8	PP-006	Prepreg		71.12µm	4.1							
6	Int5 (PWR)	CF-004	Plane	1oz	35µm		✓	6 - Int5 (PWR)	8 - Bottom L...	124.69µm	49.99	0.01%	6.9403
	Dielectric 9	Core-027	Core		228.6µm	4.5							
7	Int6 (Sign)	CF-004	Signal	1oz	35µm		✓	7 - Int6 (Sign)		224.37µm	49.99	0.02%	6.0745
	Dielectric 10	PP-006	Prepreg		71.12µm	4.1							
	Dielectric 11	PP-006	Prepreg		71.12µm	4.1							
8	Bottom Layer	CF-004	Signal	1oz	35µm								
	Bottom Surface...	PbSn	Surface Finish		20µm								
	Bottom Solder	SM-001	Solder Mask		25.4µm	4							
	Bottom Overlay		Overlay										

Ojo con la asignación de planos de referencia

Properties

Layer Stack Manager

Search

Impedance Profile

Description

Type Single

Target Impedance 50

Target Tolerance 10%

Transmission Line

Simulated with SIMBEOR® software

Use Surface Finish ☒

Use Solder Mask ☒

Trace Inverted ☐

Etch 0

Width (W1) 224.37µm

Width (W2) 224.37µm ☒

Covering (C1) 25.4µm

Covering (C2) 25.4µm

Impedance (Zo) 49.99

Deviation 0.02%

Delay (Tp) 6.07453ns/m

Inductance 303.65879nH/m

Capacitance 121.51731pF/m

IMPEDANCIA DE LÍNEA CON STACK-UP EDITOR

Stackup_practice1.PcbDoc * Stackup_practice1.PcbDoc [Stackup] *

+ Add Remove D100 **S50** Features

#	Name	Material	Type	Weight	Thickness	Dk	Top Ref	Bottom Ref	Width (W1)	Impe...	Devia...	Delay...
	Top Overlay		Overlay									
	Top Solder	SM-001	Solder Mask		25.4μm	4						
	Top Surface Fini...	PbSn	Surface Finish		20μm							
1	Top Layer	CF-004	Signal	1oz	35μm		✓	3 - Int2 (GND)	732.19μm	50.01	0.02%	6.096n...
	Dielectric 1	PP-006	Prepreg		71.12μm	4.1						
	Dielectric 2	PP-006	Prepreg		71.12μm	4.1						
2	Int1 (Sign)	CF-004	Signal	1oz	35μm		✓	3 - Int2 (GND)	333.23μm	49.98	0.04%	6.6500...
	Dielectric 3	Core-027	Core		228.6μm	4.5						
3	Int2 (GND)	CF-004	Plane	1oz	35μm							
	Dielectric 4	PP-006	Prepreg		71.12μm	4.1						
	Dielectric 5	PP-006	Prepreg		71.12μm	4.1						
4	Int3 (Sign)	CF-004	Signal	1oz	35μm		✓	3 - Int2 (GND) 5 - Int4 (Sign)	124.69μm	49.99	0.01%	6.9403...
	Dielectric 6	Core-027	Core		228.6μm	4.5						
5	Int4 (Sign)	CF-004	Signal	1oz	35μm		✓	4 - Int3 (Sign) 6 - Int5 (PWR)	124.69μm	49.99	0.01%	6.9403...
	Dielectric 7	PP-006	Prepreg		71.12μm	4.1						
	Dielectric 8	PP-006	Prepreg		71.12μm	4.1						
6	Int5 (PWR)	CF-004	Plane	1oz	35μm							
	Dielectric 9	Core-027	Core		228.6μm	4.5						
7	Int6 (Sign)	CF-004	Signal	1oz	35μm		✓	6 - Int5 (PWR)	333.23μm	49.98	0.04%	6.6500...
	Dielectric 10	PP-006	Prepreg		71.12μm	4.1						
	Dielectric 11	PP-006	Prepreg		71.12μm	4.1						
8	Bottom Layer	CF-004	Signal	1oz	35μm		✓	6 - Int5 (PWR)	732.19μm	50.01	0.02%	6.096n...
	Bottom Surface...	PbSn	Surface Finish		20μm							
	Bottom Solder	SM-001	Solder Mask		25.4μm	4						
	Bottom Overlay		Overlay									

Stackup Impedance Via Types

Properties

Layer Stack Manager

Search

Impedance Profile

Description

Type **Single**

Target Impedance 50

Target Tolerance 10%

Transmission Line

Simulated with SIMBEOR® software

Use Surface Finish ☐

Use Solder Mask ☒

Trace Inverted ☐

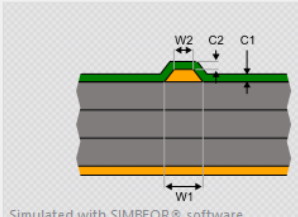
Etch 0

Width (W1) 732.19μm

Width (W2) 732.19μm **fx**

Covering (C1) 25.4μm

Covering (C2) 25.4μm



IMPEDANCIA DE LÍNEA CON STACK-UP EDITOR

+ Add

Remove

D100

Features

#	Name	Material	Type	Weight	Thickness	Dk	Top Ref	Bottom Ref	Width (W1)	Trace Gap (G)	Impedance...	Devia...	
	Top Overlay		Overlay										
	Top Solder	SM-001	Solder Mask		25.4µm	4							
	Top Surface Fini...	PbSn	Surface Finish		20µm								
1	Top Layer	CF-004	Signal	1oz	35µm		✓	3 - Int2 (GND)	152.22µm	127µm	99.98	0.02%	
	Dielectric 1	PP-006	Prepreg		71.12µm	4.1							
	Dielectric 2	PP-006	Prepreg		71.12µm	4.1							
2	Int1 (Sign)	CF-004	Signal	1oz	35µm		✓	3 - Int2 (GND)	100µm	127µm	96.41	3.59%	
	Dielectric 3	Core-027	Core		228.6µm	4.5							
3	Int2 (GND)	CF-004	Plane	1oz	35µm								
	Dielectric 4	PP-006	Prepreg		71.12µm	4.1							
	Dielectric 5	PP-006	Prepreg		71.12µm	4.1							
4	Int3 (Sign)	CF-004	Signal	1oz	35µm		✓	3 - Int2 (GND)	6 - Int5 (PWR)	100µm	127µm	89.06	10.94%
	Dielectric 6	Core-027	Core		228.6µm	4.5							
5	Int4 (Sign)	CF-004	Signal	1oz	35µm		✓	3 - Int2 (GND)	6 - Int5 (PWR)	100µm	127µm	89.06	10.94%
	Dielectric 7	PP-006	Prepreg		71.12µm	4.1							
	Dielectric 8	PP-006	Prepreg		71.12µm	4.1							
6	Int5 (PWR)	CF-004	Plane	1oz	35µm								
	Dielectric 9	Core-027	Core		228.6µm	4.5							
7	Int6 (Sign)	CF-004	Signal	1oz	35µm		✓	6 - Int5 (PWR)		100µm	127µm	96.41	3.59%
	Dielectric 10	PP-006	Prepreg		71.12µm	4.1							
	Dielectric 11	PP-006	Prepreg		71.12µm	4.1							
8	Bottom Layer	CF-004	Signal	1oz	35µm		✓	6 - Int5 (PWR)		152.22µm	127µm	99.98	0.02%
	Bottom Surface...	PbSn	Surface Finish		20µm								
	Bottom Solder	SM-001	Solder Mask		25.4µm	4							
	Bottom Overlay		Overlay										

Stackup Impedance Via Types

Layer Stack Manager

Search

Impedance Profile

Description

Type Differential

Target Impedance 100

Target Tolerance 10%

Transmission Line

Simulated with SIMBEOR® software

Trace Inverted

Etch 0

Width (W1) 100µm

Width (W2) 100µm

Trace Gap (G) 127µm

Impedance (Zdiff) 96.41

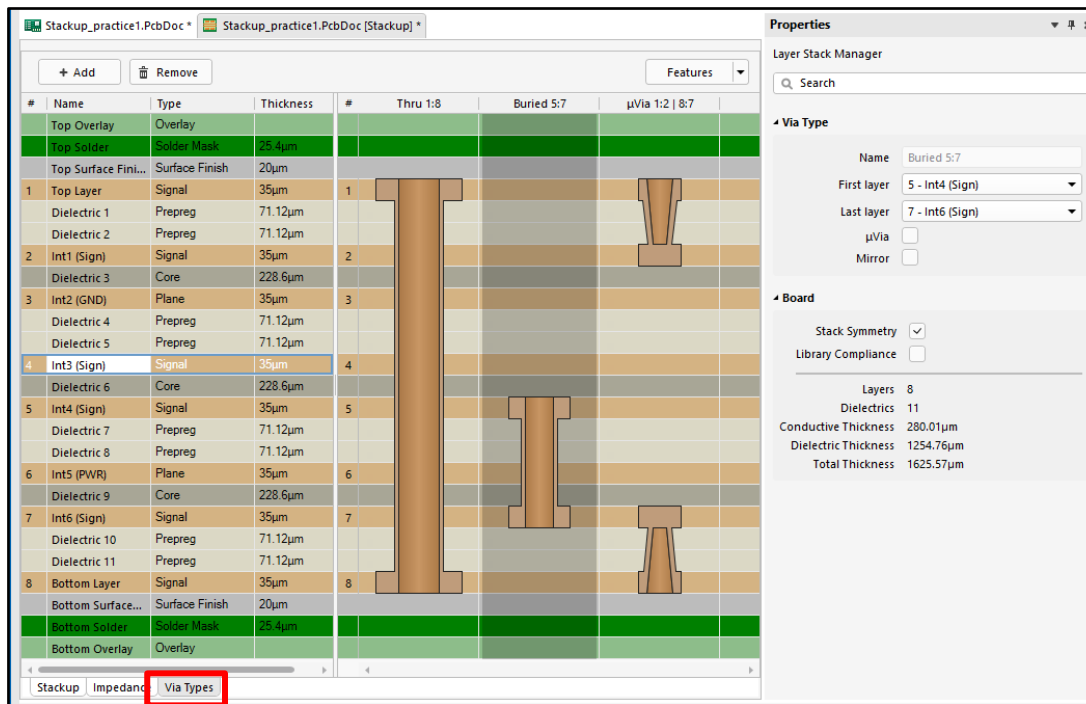
Deviation 3.59%

Delay (Tp) 6.82422ns/m

Inductance 657.90362nH/m

Capacitance 70.79540nF/m

IMPEDANCIA DE LÍNEA CON STACK-UP EDITOR

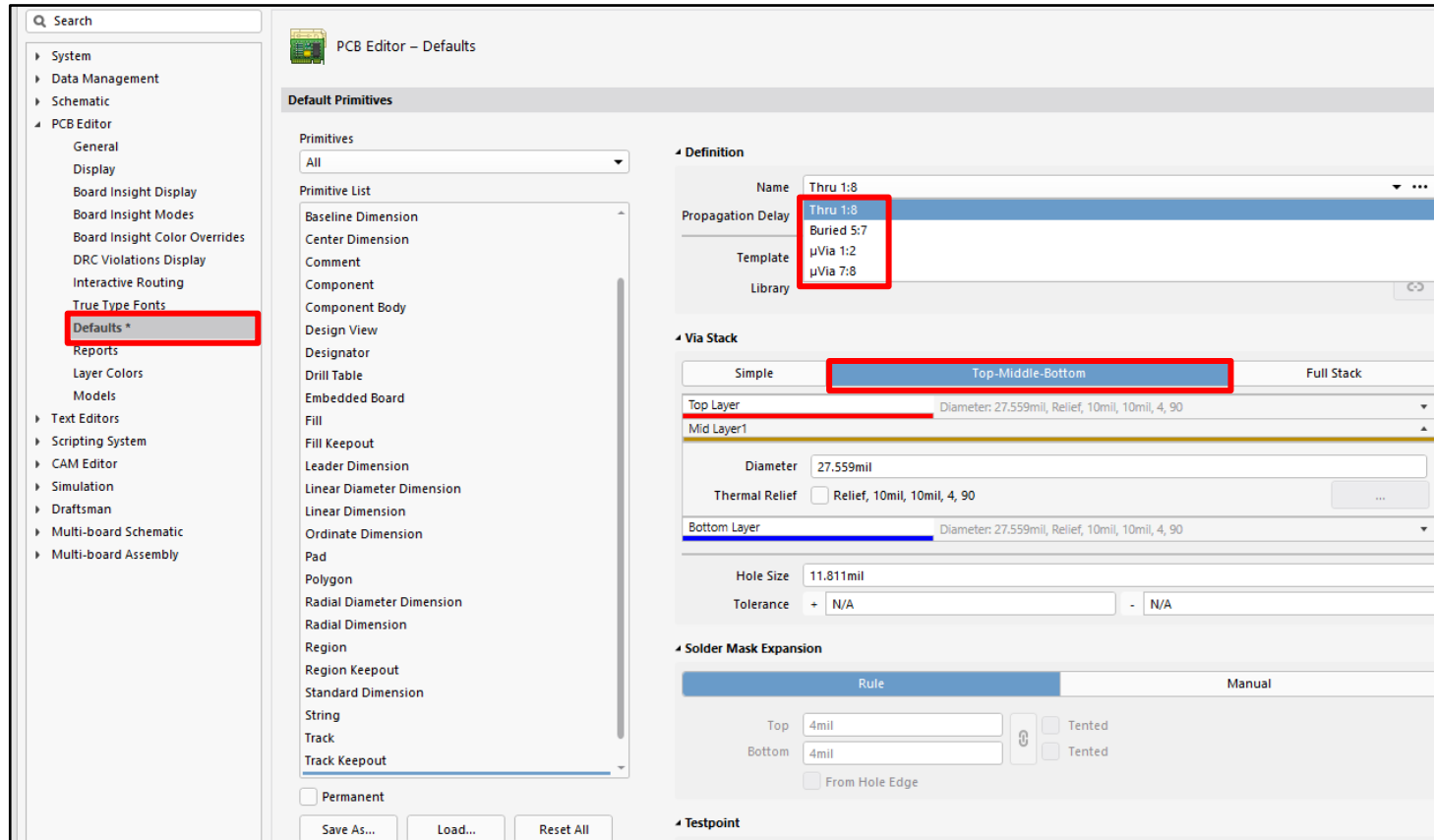


En la pestaña **Via Types** añadimos los tipos de vías que usaremos en el diseño.

Aquí sólo definimos la dimensión z.

El resto de propiedades se definen mediante reglas, edición manual o “PCB default primitives”

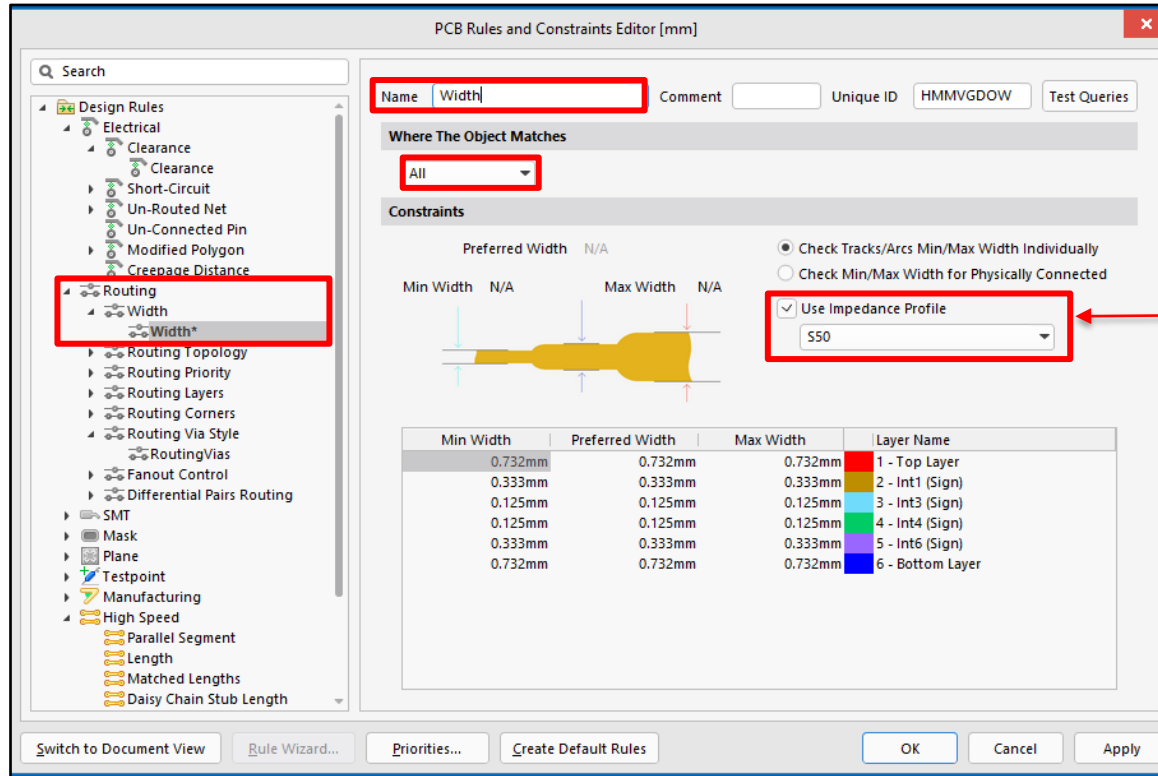
IMPEDANCIA DE LÍNEA CON STACK-UP EDITOR



CONTENIDOS

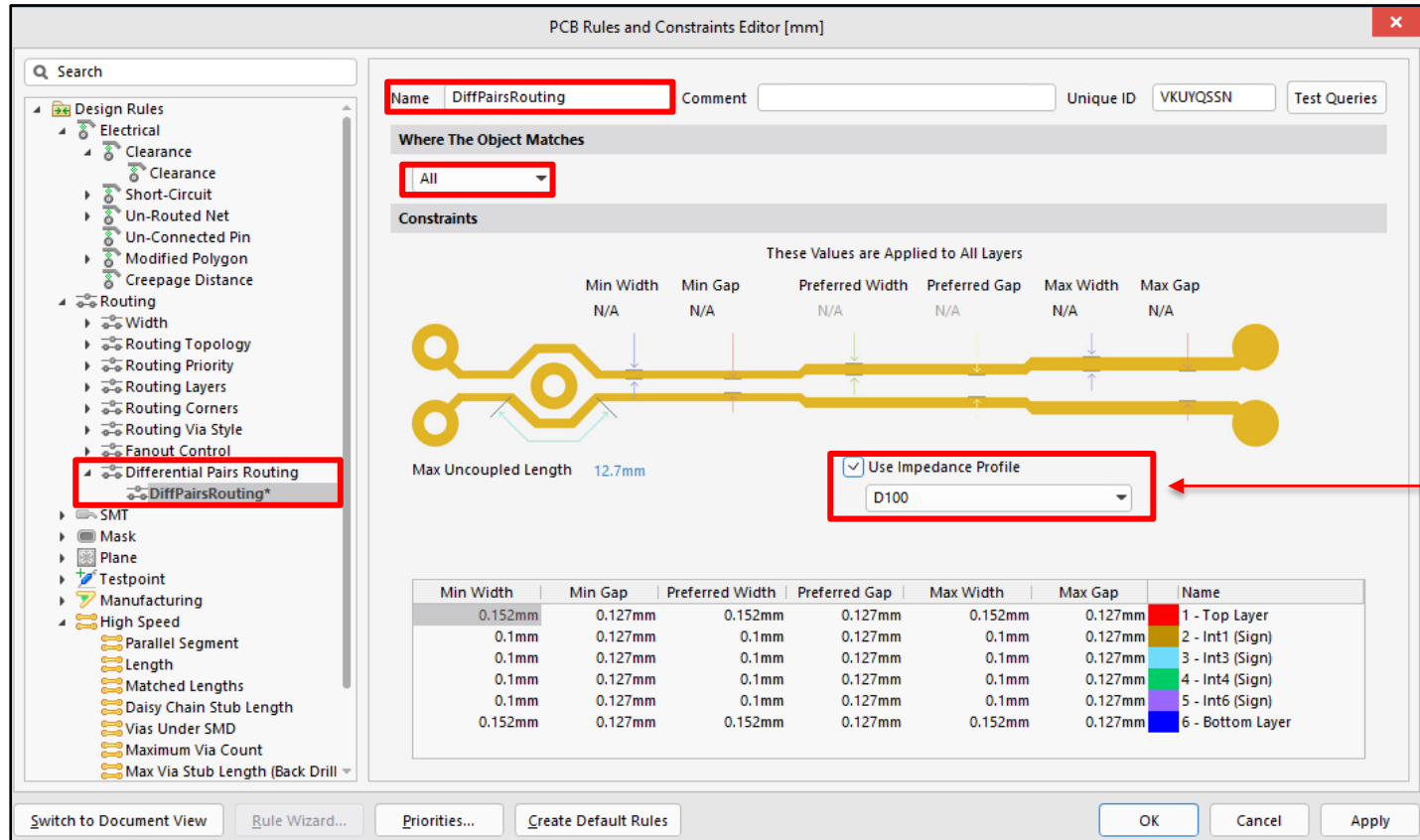
- Impedancia de línea con stack-up editor
- Reglas de diseño básicas
- Reglas de diseño *High Speed*
- Reglas de diseño: directivas en esquemas

REGLAS DE DISEÑO BÁSICAS



Carga anchuras de pista desde el stack-up editor

REGLS DE DISEÑO BÁSICAS

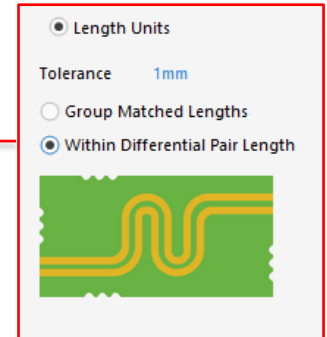
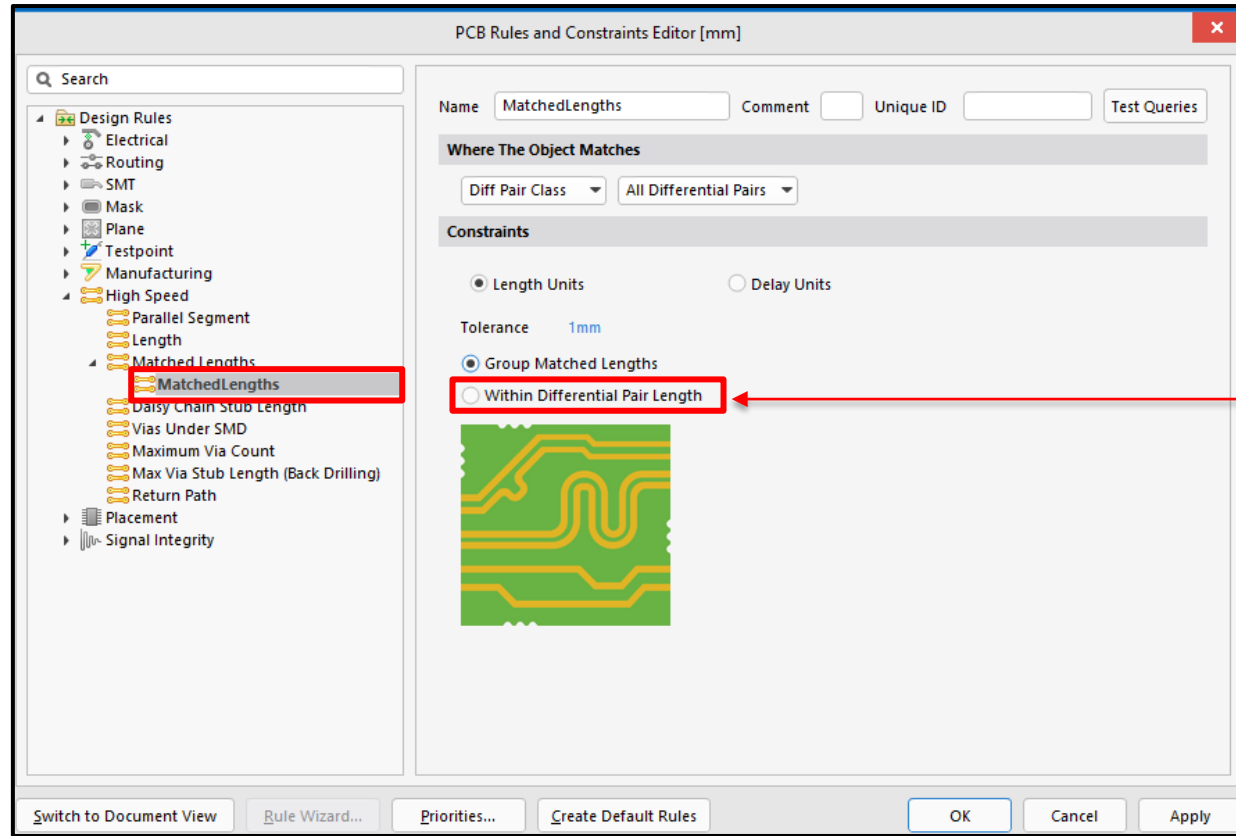


Carga W, S desde el stack-up editor

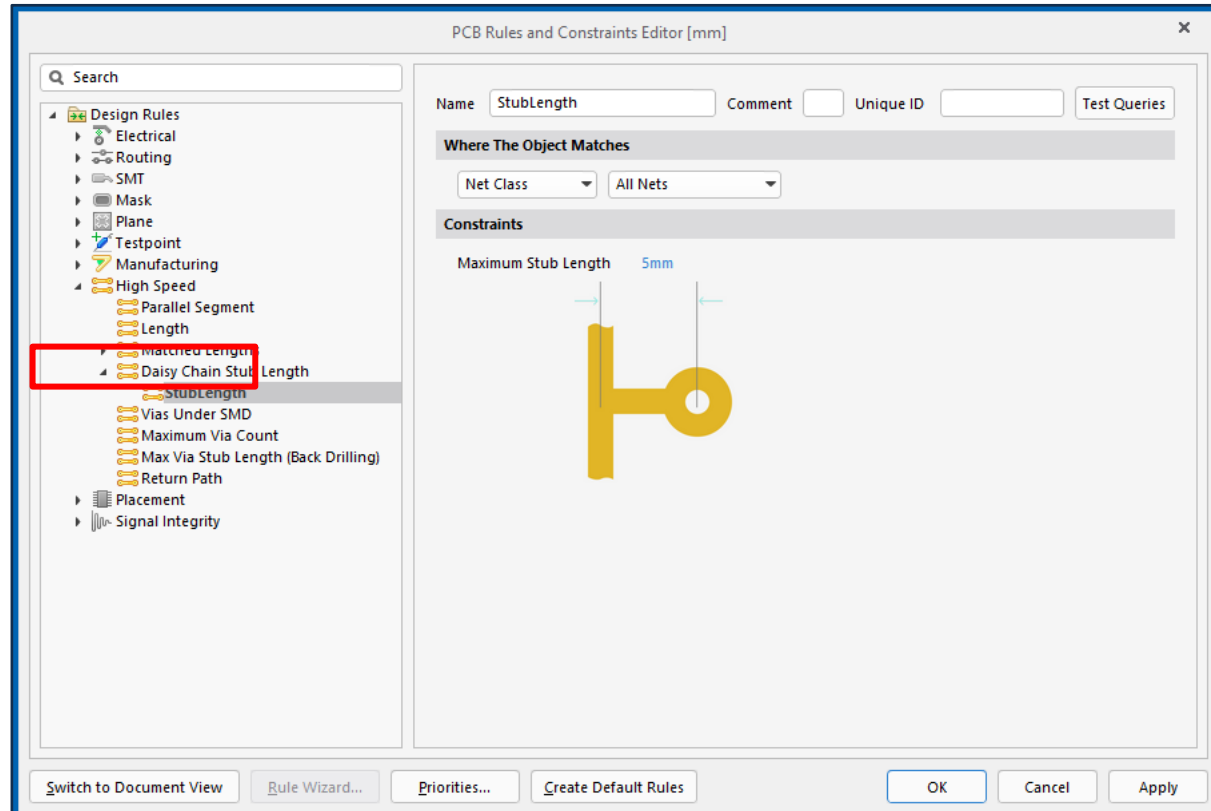
CONTENIDOS

- Impedancia de línea con stack-up editor
- Reglas de diseño básicas
- Reglas de diseño *High Speed*
- Reglas de diseño: directivas en esquemas

REGLAS DE DISEÑO HIGH SPEED

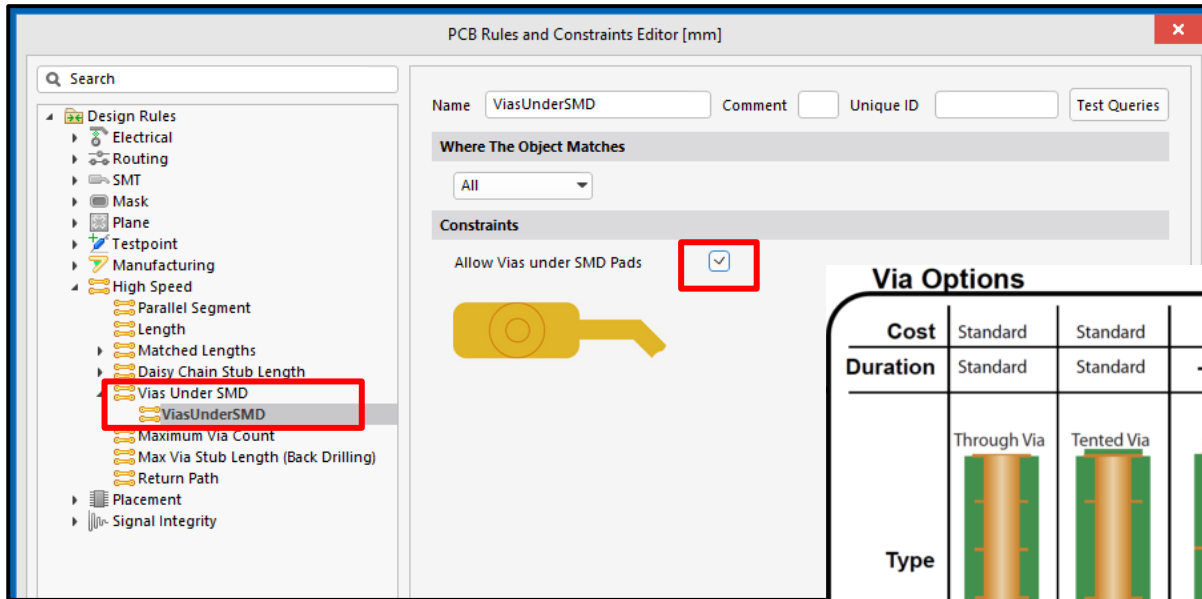


REGLAS DE DISEÑO HIGH SPEED



REGLAS DE DISEÑO HIGH SPEED

Permite usar tecnología via-in-pad

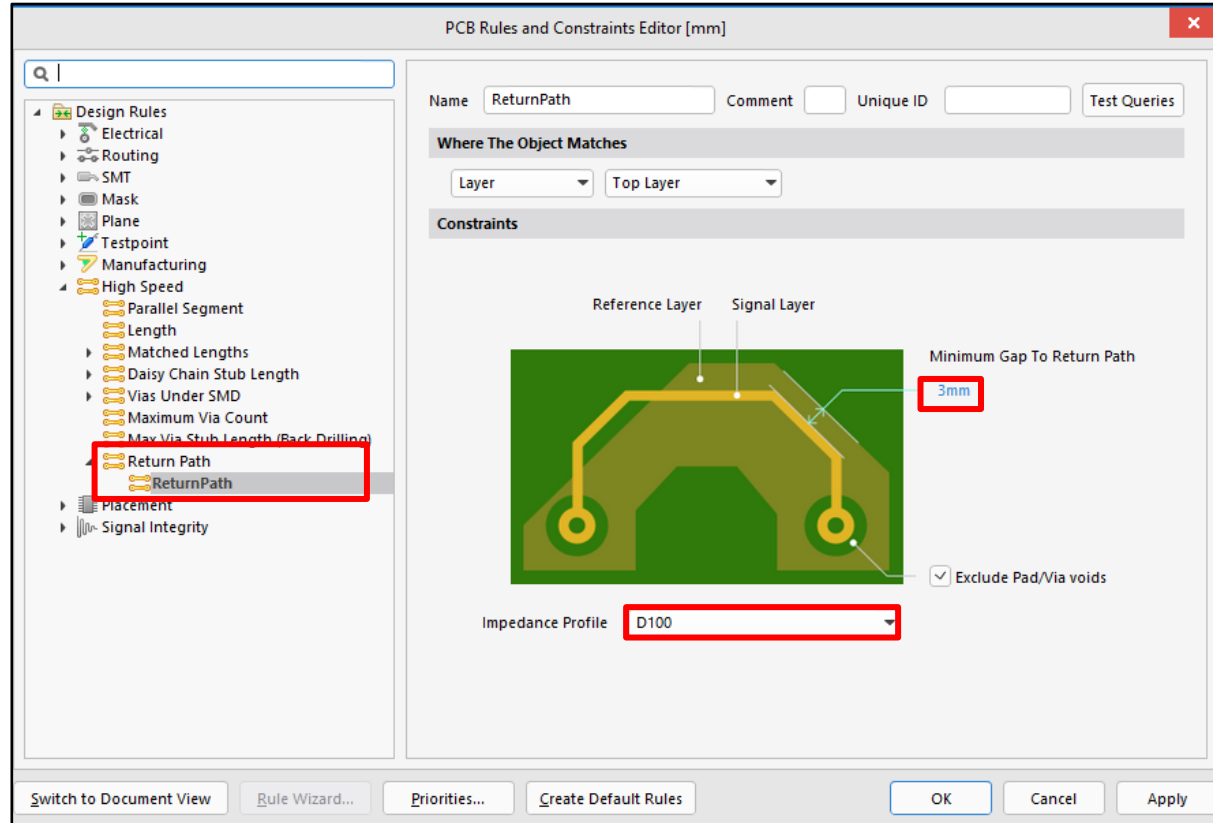


Via Options						
Cost	Standard	Standard	+\$	++	+++	+++
Duration	Standard	Standard	+🕒	+🕒	+🕒🕒	+🕒🕒
Type	Through Via	Tented Via	Blind Via	Buried Via	Stacked Via	Via in Pad
						

Via-in-pad requiere rellenar y cubrir vías. ¿Lo permite tu fabricante de PCBs?

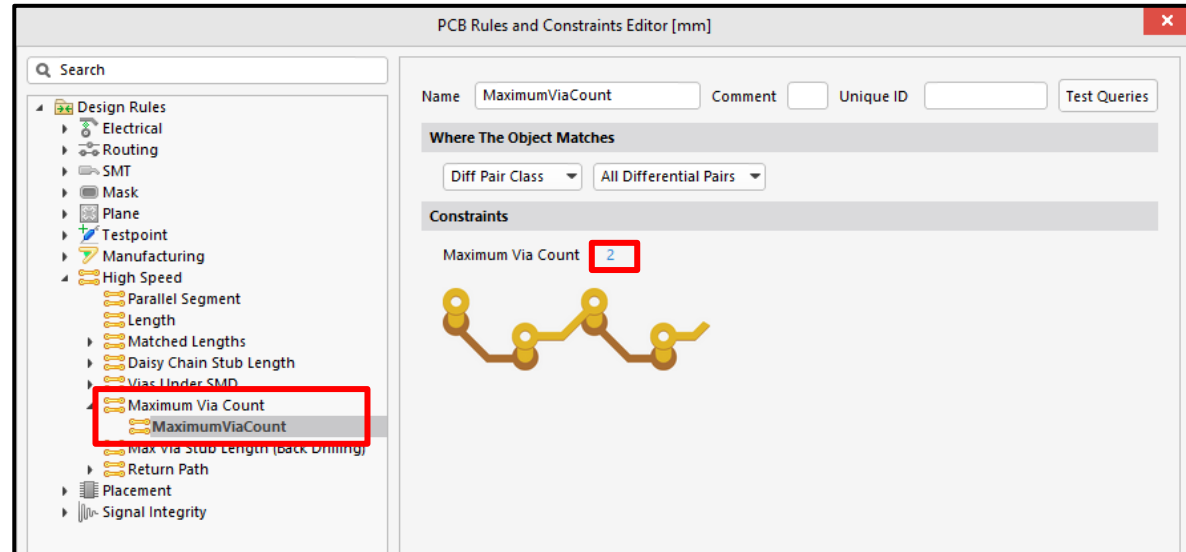
REGLAS DE DISEÑO HIGH SPEED

Asegura la continuidad de los caminos de retorno



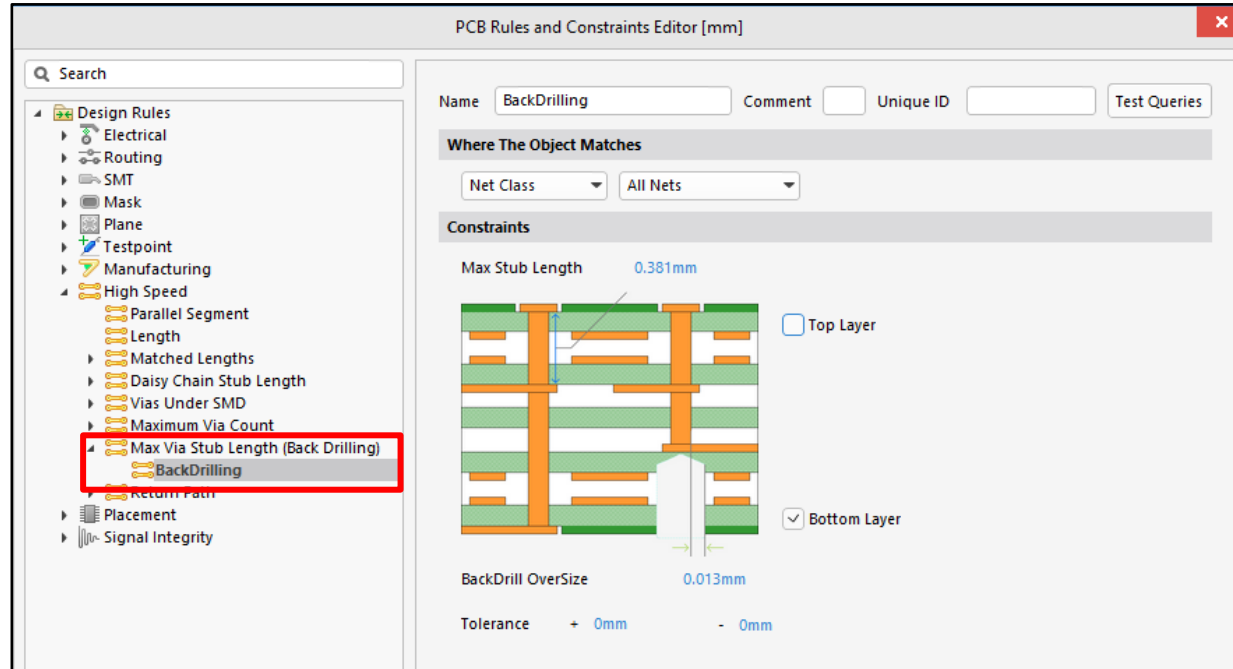
REGLAS DE DISEÑO HIGH SPEED

Útil para limitar el número de transiciones en enlaces como SATA y PCIe



Ejemplos de recomendaciones: no pasar de 4 vías (USB 2.0), 2 vías (USB3), 0-2 vías (PCIe, SATA)

REGLAS DE DISEÑO HIGH SPEED

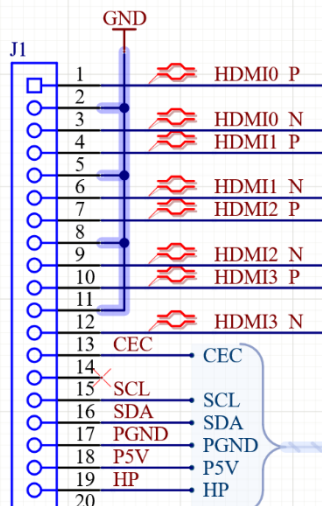


El proceso de *back-drilling* encarece el PCB. Intenta evitar los stubs mediante un adecuado rutado/stack-up

CONTENIDOS

- Impedancia de línea con stack-up editor
- Reglas de diseño básicas
- Reglas de diseño *High Speed*
- Reglas de diseño: directivas en esquemas

REGLAS DE DISEÑO: DIRECTIVAS EN LOS ESQUEMAS

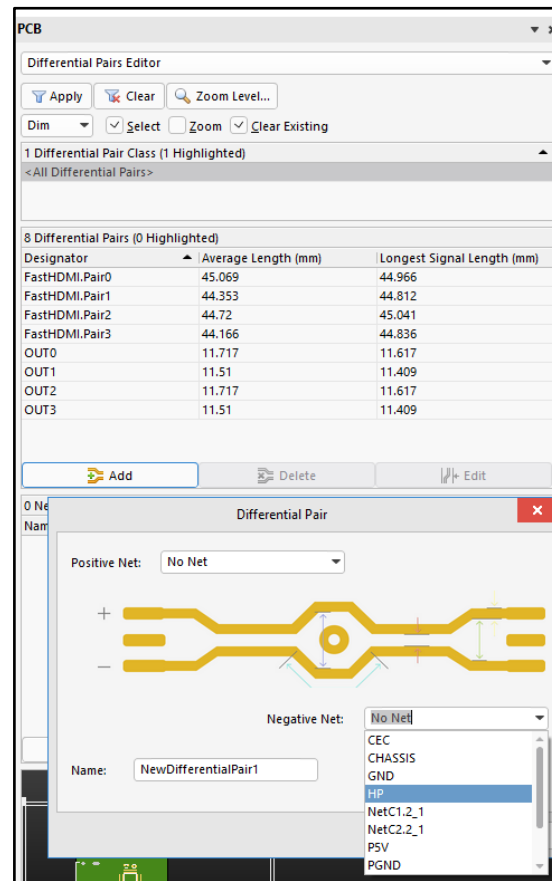


Los pares diferenciales en Altium requieren:

(1) dos señales nombradas “_P” y “_N”,

(2) Una **directiva** “differential pair” en el esquema o mediante “Differential pairs editor” en **PCB panel**

Recomendación: hacerlo en el esquema, ya que añade claridad e información de cara al ingeniero/a que realice el PCB



REGLAS DE DISEÑO: DIRECTIVAS EN LOS ESQUEMAS

Uso de Parameter Set y Blankets para definir reglas de diseño a un conjunto de nets

The screenshot displays the Altium Designer interface for defining design rules. On the left, the 'Choose Design Rule Type' dialog is open, showing a tree structure of rule categories. The 'Signal Integrity' category is expanded, and 'Matched Net Lengths' is selected, indicated by a red '3'. In the center, a schematic diagram shows a vertical bus of 16 nets. A red dashed box labeled '1 Parameter Set' encloses the first three nets (OUT0 P, OUT0 N, OUT1 P). A red '2' is placed at the bottom right of the schematic. On the right, the 'Properties' panel is shown, with the 'Parameter Set' tab selected. The 'Label' is set to 'Parameter Set' and the 'Style' is 'Large'. The 'Parameters' section shows 'No Parameters', 'No Rules', and 'No Classes'.

Choose Design Rule Type

- Solder Mask Expansion
- Paste Mask Expansion
- Plane
 - Power Plane Connect Style
 - Power Plane Clearance
 - Polygon Connect Style
- Testpoint
 - Fabrication Testpoint Style
 - Fabrication Testpoint Usage
 - Assembly Testpoint Style
 - Assembly Testpoint Usage
- Manufacturing
 - Minimum Annular Ring
 - Acute Angle Constraint
 - Hole Size Constraint
 - Layer Pairs
- High Speed
 - Parallel Segment Constraint
 - Length Constraint
 - Matched Net Lengths
 - Daisy Chain Stub Length
 - Vias Under SMD Constraint
 - Maximum Via Count Constraint
- Placement
 - Component Clearance Constraint
 - Component Orientations Rule
 - Permitted Layers Rule
 - Nets to Ignore
 - Height Constraint
- Signal Integrity
 - Signal Stimulus
 - Overshoot - Falling Edge
 - Overshoot - Rising Edge
 - Undershoot - Falling Edge
 - Undershoot - Rising Edge
 - Impedance Constraint
 - Signal Top Value

Properties

Parameter Set Components (and 11 more)

Search

Location

(X/Y) 226.06mm 147.32mm

Rotation 90 Degrees

Properties

Label Parameter Set

Style Large

Parameters

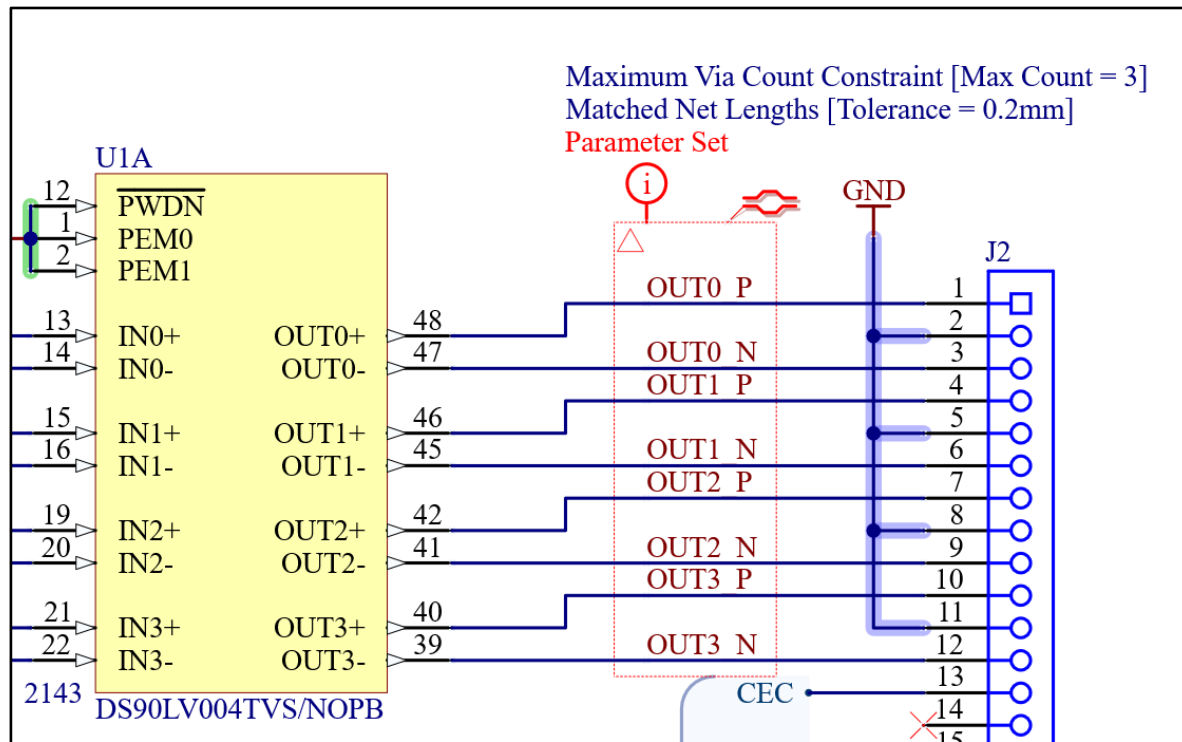
All Parameters Rules Classes

Name	Value
No Parameters	
No Rules	
No Classes	

REGLAS DE DISEÑO: DIRECTIVAS EN LOS ESQUEMAS

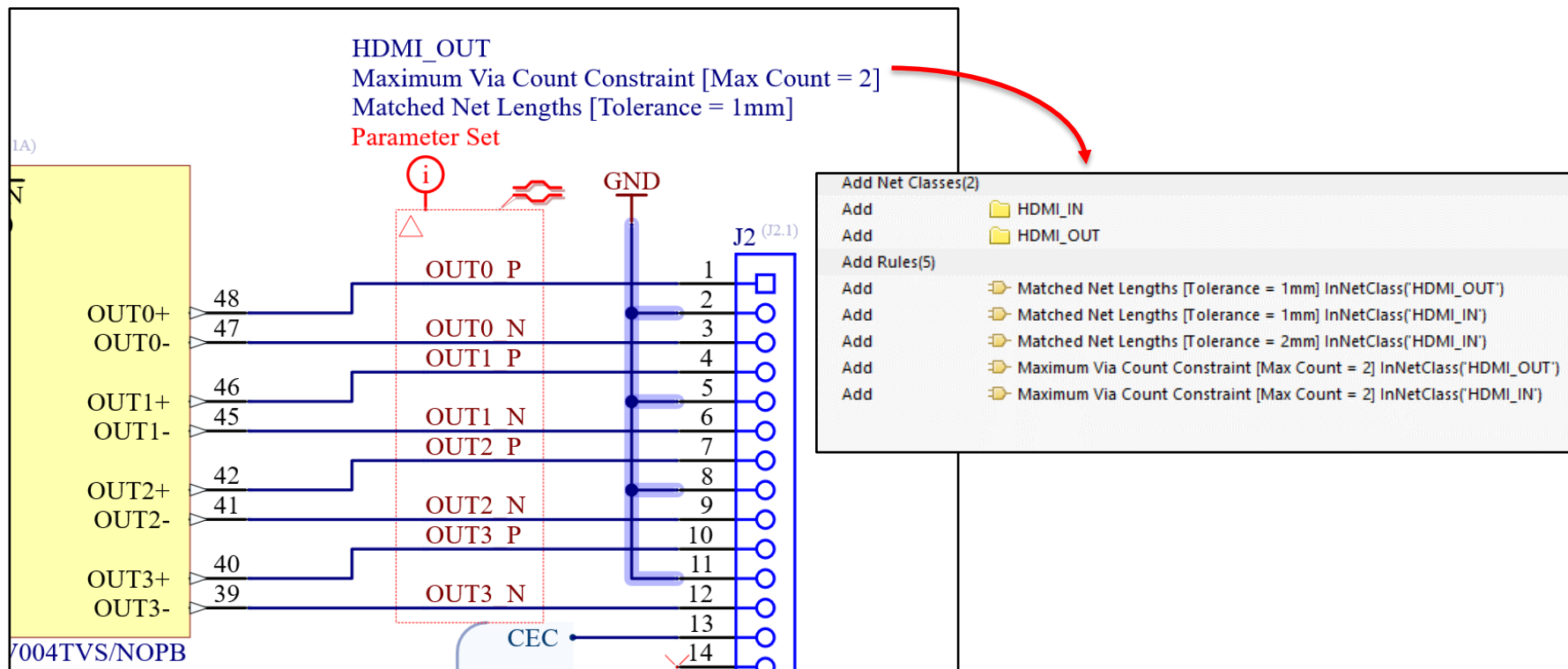
Hemos definido 4 pares diferenciales y dos reglas de diseño (*max via count*, *matched net lengths*)

Mejora claridad, documentación y facilita revisión. **PERO** se crea una regla por cada *net*



REGLAS DE DISEÑO: DIRECTIVAS EN LOS ESQUEMAS

Para evitar crear una regla por net, define una clase (HDMI_OUT) en la misma directiva



REGLAS DE DISEÑO: DIRECTIVAS EN LOS ESQUEMAS

Revisa que la revisión de las reglas *High Speed* está activada (*Online y High Speed*)

